

# BreathWise

## AI Chest X-Ray Analysis and Clinical Decision Support System

### University of Sadat City

Faculty of Computers and Artificial Intelligence

Department of Information Systems — Academic Year 2026

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# The Problem

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## Radiologist Shortage & Burnout

Imaging volume growing faster than radiologist training.  
Cognitive fatigue → diagnostic errors.



## Diagnostic Delay in Emergencies

Hours of delay detecting Pneumothorax in ER/ICU can be fatal.



## Inter-Observer Variability

Same X-ray, different diagnoses depending on the reader and experience level.



## The 'Black Box' Barrier

Clinicians distrust AI that can't visually explain its reasoning.

# Disease Burden — Why These 4 Conditions?

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**Pneumonia**

**450M  
cases/yr**

Leading infectious killer in children under 5 (740K deaths/yr)

**Cardiomegaly**

**64M  
cases/yr**

Heart failure, cardiomyopathy — high mortality & hospitalization

**Pleural Effusion**

**10M  
cases/yr**

Secondary to heart failure, pneumonia, and malignancies

**Pneumothorax**

**0.5M  
cases/yr**

Acute life-threatening emergency requiring immediate intervention

# Project Objectives

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## High-Accuracy Classification

Multi-label prediction with >90% accuracy across all thoracic conditions



## Near Real-Time Inference

<0.5s per scan (single pass), ~2s with Test-Time Augmentation



## Explainability by Design

Grad-CAM heatmaps overlaying every prediction to break the 'black box'



## Clinical Workspace

Patient tracking, scan histories, automated PDF reports with QuestPDF

# System Architecture (3-Tier Microservices)

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## React Frontend

Port 5173

TypeScript, Vite, TailwindCSS  
Recharts, Axios

## ASP.NET Core 8 Backend

Port 5169

C#, EF Core, SQL Server  
JWT + RBAC, QuestPDF

## FastAPI / PyTorch AI Service

Port 8000

Python, DenseNet-121  
Grad-CAM, OpenCV

Backend acts as secure proxy → AI service never exposed to public internet. Internal auth: Authorization: internal-key

# Technology Stack

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## Frontend

- React 19
- TypeScript
- Vite
- TailwindCSS
- Recharts
- Axios

## Backend

- C# .NET 8
- ASP.NET Core 8
- Entity Framework Core
- SQL Server
- QuestPDF
- JWT + RBAC

## AI Service

- Python 3.12
- FastAPI + Uvicorn
- PyTorch (DenseNet-121)
- OpenCV + PIL
- torchvision
- Grad-CAM

# End-to-End Data Flow

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1

## Upload

Doctor uploads X-ray  
via React drag-and-  
drop

2

## Save

C# saves image as  
wwwroot/uploads/[GUI  
D].png

3

## AI Call

C# sends internal  
POST  
to  
localhost:8000/predict

4

## Inference

Python runs  
DenseNet-121  
+ TTA (3 passes) +  
Grad-CAM

5

## DB Write

C# decodes base64  
heatmap  
writes ScanResult to  
SQL

6

## Display

React renders  
diagnosis  
+ interactive heatmap  
toggle

# The AI Model — DenseNet-121

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## Dense Connectivity

Each layer receives feature maps from ALL preceding layers

Growth Rate  $k = 32$  (new feature maps per layer)

Transition Layers:  $1 \times 1$  conv +  $2 \times 2$  avg pool between blocks

## Classifier Head

`nn.Linear(1024, 5)` — 5 output logits

Loss: `BCEWithLogitsLoss` (Binary Cross-Entropy)

Optimizer: Adam, LR =  $1e-4$ , Batch Size = 32

## Why DenseNet over ResNet?

1024 output features (vs ResNet's 2048) → more compact

Better gradient flow through all layers



# Explainability — Grad-CAM

## Breaking the Black Box

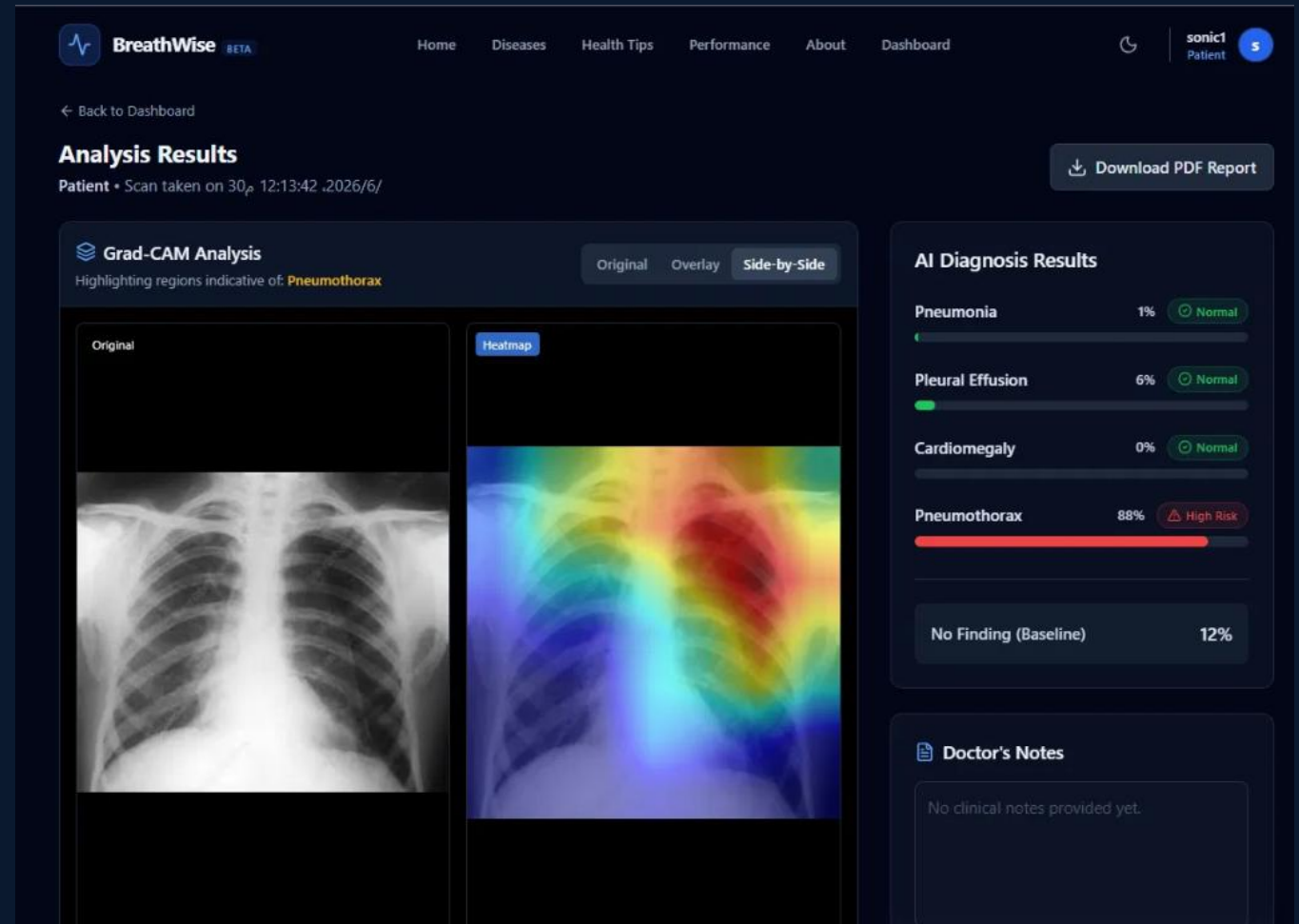
Computes gradients of class score vs DenseBlock4 activations

Global Average Pooling → importance weights  $\alpha$

Weighted sum → ReLU → upsampled to 224×224

Mapped to JET color palette, blended  $\alpha=0.4$

3 View Modes: Original | Overlay | Side-by-Side



# Dataset & Class Balancing

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## Original NIH ChestX-ray14

112,120 images — 30,805 patients

15 finding labels (14 pathologies + No Finding)

Heavily imbalanced: No Finding = 60,361 (53.8%)

Rarest: Hernia = 227 (0.2%)

Problem: Training on raw data → biased model that always predicts 'Healthy'

## Filtered & Balanced (BreathWise)

Filtered to 5 target conditions

Balanced to 5,500 per class

Total: 27,500 images

Train: 22,000 (80%) / Val: 5,500 (20%)

Pneumonia (1,431 → 5,500) via oversampling with replacement

# Training & Results

## Training Results & Statistics

Detailed performance metrics of our core DenseNet-121 model. We trained the network on thousands of chest X-rays to achieve high accuracy and reliable screening results.



Best Validation Loss

0.2668



Overall Accuracy

89.67%



Training Epochs

5

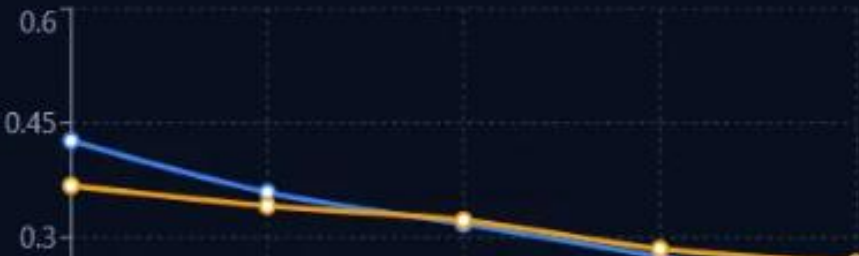


Inference Time (TTA)

~2s

### Training Progression

Training & Validation Loss



Loss Reduction Over Epochs



# Per-Class Performance (Classification Report)

| Condition        | Precision | Recall | F1-Score | Support |
|------------------|-----------|--------|----------|---------|
| Cardiomegaly ★   | 0.82      | 0.85   | 0.84     | 1,250   |
| Pneumonia        | 0.85      | 0.77   | 0.81     | 1,148   |
| Pneumothorax     | 0.79      | 0.76   | 0.77     | 1,194   |
| Pleural Effusion | 0.79      | 0.72   | 0.75     | 1,928   |
| No Finding       | 0.73      | 0.66   | 0.69     | 1,113   |
| Macro Avg        | 0.80      | 0.75   | 0.77     | 6,633   |

★ Cardiomegaly = highest F1 (0.84) due to distinct visual signature of enlarged heart  
No Finding = lowest F1 (0.69) — expected, as it represents absence of features. Mitigated by mathematical derivation in production.

# Post-Training Optimizations

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## Optimal Threshold Tuning

Default 0.5 → sweep 0.0–1.0, maximize F1 per class

Pneumonia: 0.39

Effusion: 0.45

Cardiomegaly: 0.54

Pneumothorax: 0.39

No Finding: 0.43

Acute conditions get lower thresholds → higher sensitivity → fewer missed diagnoses

## Test-Time Augmentation (TTA)

3 predictions per image:

Original + H-Flip + 10° Rotation

Final = arithmetic mean of 3 sigmoid outputs

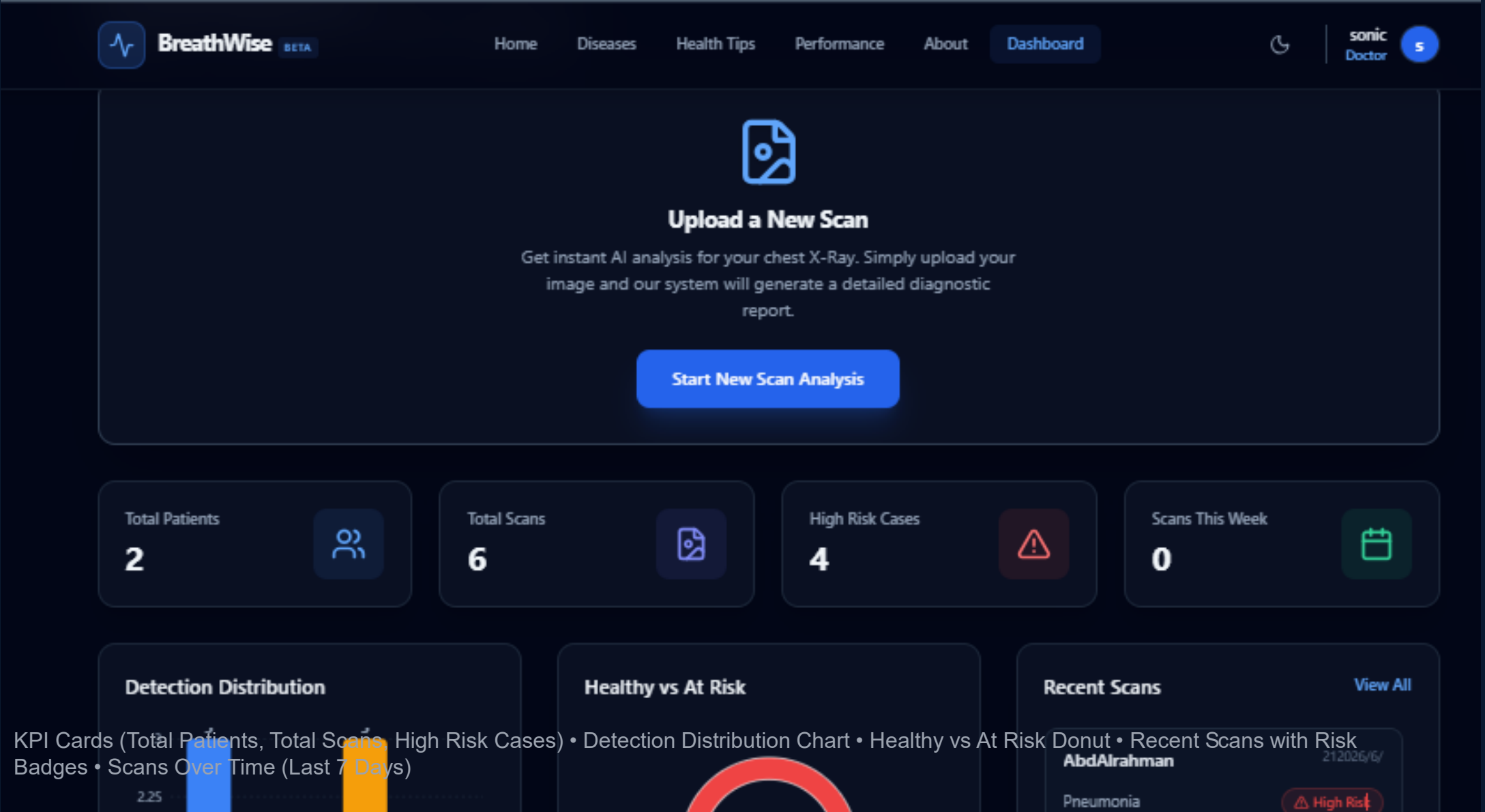
## Progressive Accuracy:

Baseline (0.5): 89.37%

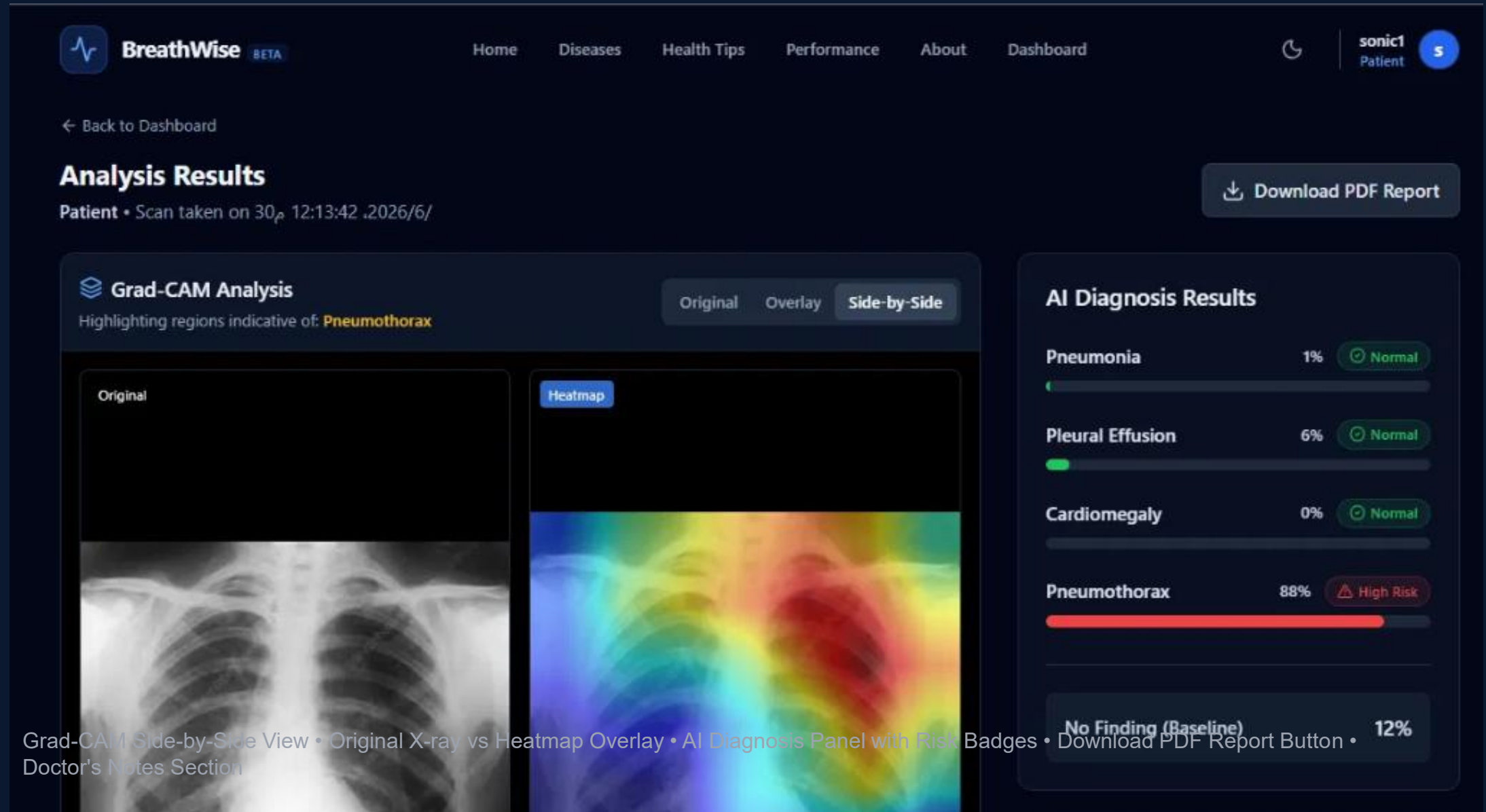
+ Thresholds: 89.33%

+ TTA: 89.67% ✓

# Doctor Dashboard — Clinical Workspace



# AI Diagnosis & Grad-CAM Results



Grad-CAM Side-by-Side View • Original X-ray vs Heatmap Overlay • AI Diagnosis Panel with Risk Badges • Download PDF Report Button • Doctor's Notes Section

# Automated Clinical

## PDF Reports

QuestPDF Report Engine

Professional radiology report layout

Patient demographics & study info

AI Predictive Analysis with probability bars

Overall Risk Severity Gauge (Low / Moderate / High)

Probability Distribution Chart

Clinical Impression with recommendations

Attached X-ray + Grad-CAM heatmap images

Generated via C# QuestPDF — instant compilation

Requires clinician's electronic signature (human-in-the-loop)

### BreathWise Diagnostics Center

123 Health Avenue, Medical District  
Tel: +1 (555) 019-8372 | Email: radiology@BreathWise.com

### RADIOLOGY REPORT

Date: Jun 30, 2026  
Report ID: 7E3D578E

**Patient Name:** AbdAlrahman  
**DOB / Age:** 2026-06-02 (0 yrs)  
**Gender:** Male

**Study:** XR Chest PA/AP  
**Study Date:** Jun 21, 2026 10:50  
**Referring Dr:** sonic

#### CLINICAL INDICATION:

Routine screening / AI pre-read evaluation.

#### AI PREDICTIVE ANALYSIS:

The chest radiograph was analyzed using the BreathWise deep learning algorithm. The following probability scores indicate the likelihood of specific pathophysiological findings:

|                  |                        |       |
|------------------|------------------------|-------|
| No Finding       | <div><div></div></div> | 0.0%  |
| Pneumonia        | <div><div></div></div> | 95.9% |
| Pleural Effusion | <div><div></div></div> | 5.5%  |
| Cardiomegaly     | <div><div></div></div> | 14.9% |
| Pneumothorax     | <div><div></div></div> | 12.0% |

#### Overall Risk Severity Gauge:



#### Probability Distribution Chart:



#### IMPRESSION:

- The AI algorithm indicates a high probability (95.9%) of Pneumonia. Clinical correlation is strongly recommended.
- Please refer to the Grad-CAM heatmap (Fig 2) for localization of the suspected pathophysiological regions.
- Other evaluated conditions demonstrated lower predictive probabilities.

#### ATTACHED IMAGES:



# Security & Regulatory Compliance

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## JWT Authentication

24-hour token expiry, validated on every API request



## Role-Based Access (RBAC)

Doctor (full CRUD), Patient (read-only own data), Admin



## PBKDF2 Password Hashing

HMAC-SHA256 + unique 128-bit salt per user, 10,000 iterations



## Anonymized GUID Filenames

X-rays saved as 8c4e2a87-19db-4382.png — no patient names on disk



## HIPAA/GDPR Alignment

Query-level isolation matches JWT identity claims. 17 secured API endpoints

# Reliability — Graceful Degradation

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1

## AI Service DOWN

Network timeout, crash,  
or GPU unavailable

2

## C# Catch Block

AiService catches exception  
→ falls back to GetMockScores()

3

## UI Uninterrupted

Doctors can still log in,  
view patients, browse dashboard

Also: Python AI service has its own demo mode — if densenet121.pt weights are missing, it serves mock predictions so frontend testing can proceed.

# Limitations & Future Roadmap

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## Current Limitations

- ✗ Dataset scope — NIH only, may vary on different equipment
- ✗ JPEG/PNG input only — no native DICOM support
- ✗ Single modality — chest X-rays only (no CT/MRI)
- ✗ No real-time multi-physician collaboration
- ✗ On-premise deployment only

## Future Roadmap

- DICOM & PACS Integration (fo-dicom C-STORE)
- VLM Report Writing (LLaVA-Med / Med-PaLM)
- More Pathologies: TB (1.3M deaths/yr), Atelectasis, COPD
- Federated Learning across hospitals
- Cloud-Native auto-scaling deployment

# Patient Education — Disease Knowledge Base

BreathWise BETA

[Home](#)[Diseases](#)[Health Tips](#)[Performance](#)[About](#)[Dashboard](#)



sonic1Patient

Pneumonia High

Infection that inflames air sacs in one or both lungs.

Pneumonia is an infection that inflames the air sacs (alveoli) in one or both lungs. The air sacs may fill with fluid or pus, causing cough with phlegm, fever, chills, and difficulty breathing. It can range from mild to life-threatening and is most serious for infants, young children, older adults, and people with weakened immune systems.

### Common Symptoms

- Chest pain when breathing or coughing
- Persistent cough with phlegm or pus
- Fever, sweating, and shaking chills
- Shortness of breath during normal activities
- Fatigue, nausea, vomiting, or diarrhea

### Common Causes

- Bacteria (most common: *Streptococcus pneumoniae*)
- Viruses including influenza and COVID-19
- Fungi, especially in immunocompromised patients
- Aspiration of food, drink, or saliva into lungs

### When to See a Doctor

Seek immediate medical attention if you have difficulty breathing, chest pain, persistent high fever (102°F / 39°C or higher), or a persistent cough, especially if you are coughing up pus.

Comprehensive disease information • Common symptoms & causes • When to See a Doctor alerts • Patient-accessible (read-only role)

# BreathWise

AI won't replace radiologists, but radiologists who use AI will replace those who don't.

- ✓ Multi-label classification (AUC 0.91–0.96)
- ✓ Explainable AI — Grad-CAM in production
- ✓ Secure RBAC on all 17 API endpoints
- ✓ Automated PDF reports (QuestPDF)
- ✓ Real-time dashboard (7 live metrics)
- ✓ Graceful degradation & demo mode
- ✓ HIPAA-aligned privacy (GUID, PBKDF2, JWT)
- ✓ Modular, extensible REST architecture

## Thank You!

Prof. Engy Elshafey & Prof. Asmaa Saad